H1B.48 **Motor storage property shield word 2**

Hexadecimal:	201B-31h	Effective Time:	Upon the next power-on
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	At stop

Value Range:

0 to 65535

Description

-

4.19 H30 Servo status variables read through communication

H30.00 **Servo status read through communication**

Hexadecimal:	2030-01h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

H30.00 value is hexadecimal, and is not displayed on the keypad. It is read as binary, and each bit of the binary is defined as follows:

bit	Servo State	Remarks
0	Servo ready	It determines whether the servo main circuit DC bus voltage is ready and the servo drive is ready for running. 0: Servo drive not ready 1: Servo ready
1–11	Reserved	-
12–13	Servo running state	It determines the servo running state. 00: Servo drive not ready (main circuit DC bus voltage not set up correctly) 01: Servo drive ready (main circuit DC bus voltage set up correctly, servo drive is ready for running) 10: Servo drive running (S-ON active) 11: Servo drive fault (a level 1 or level 2 fault occurs)
14–15	Reserved	-

H30.01 DO function state 1 read through communication

Hexadecimal:	2030-02h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

Used to read the state of DO functions 1 to 16 through communication. H30.01 is a hexadecimal which is not displayed on the keypad and must be converted to a binary equivalent when it is being read through communication.

bit	DO Function	Remarks
0	DO function 1 (FunOUT.1: S-RDY, servo ready)	0: Servo drive not ready 1: Servo ready
...		
15	DO function 16 (FunOUT.16: HomeAttain, homing output)	0: Home not found 1: Home found

H30.02 DO function state 2 read through communication

Hexadecimal:	2030-03h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

Bit 0 corresponds to DO function 17.

Bit 1 corresponds to DO function 18.

Bit 2 corresponds to DO function 19.

…

By analogy

bit	DO Function	Remarks
0	DO function 17 (FunOUT.17: S-ElecHomeAttain, electrical homing output)	0: Electrical homing not completed 1: Electrical homing completed
...		
4 to 15	Reserved	-

H30.03 Input pulse reference sampling value read through communication

Hexadecimal:	2030-04h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H30.04 DI status read through communication

Hexadecimal:	2030-05h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

4.20 H31 Related variables set through communication

H31.00 VDI virtual level set through communication

Hexadecimal:	2031-01h	Effective Time:	Real time
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Immediately

Value Range:

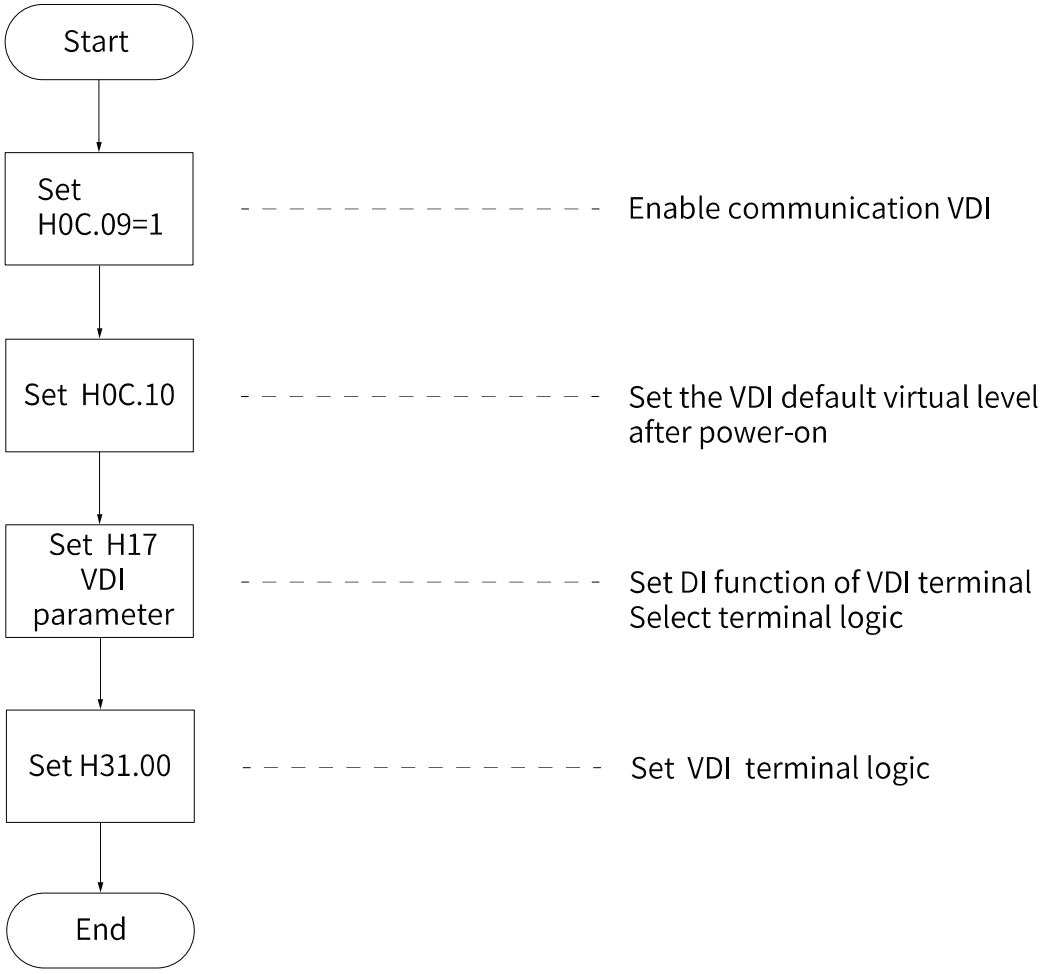
0–65535

Description

When H0C.09 is set to 1, the VDI state is defined by H31.00.

The VDI logic is determined by H0C.10 (Default VDI virtual level value upon power-on) upon initial power-on. Then, H31.00 is determined by the VDI logic.

"bit(n) = 1" of H31.00 indicates the logic of VDI (n+1) is "1". "bit(n)=0" indicates the logic of VDI (n+1) is "0".
Use the VDI according to the following procedure:



H31.04 DO state set through communication

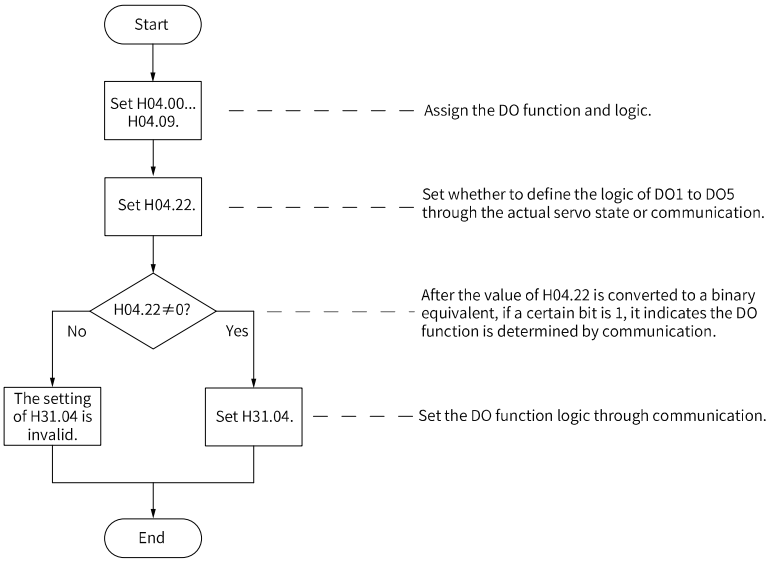
Hexadecimal:	2031-05h	Effective Time:	Real time
Min.:	0	Unit:	-
Max.:	31	Data Type:	UInt16
Default:	0	Change:	Immediately

Value Range:

0 to 31

Description

Set H04.22 to define the DO state source by H31.04.
Use the DO according to the following procedure:



H31.09 Speed reference set through communication

Hexadecimal:	2031-0Ah	Effective Time:	Real time
Min.:	-6000.000	Unit:	rpm
Max.:	6000.000	Data Type:	Int32
Default:	0.000	Change:	Immediately

Value Range:

-6000.000rpm to 6000.000rpm

Description

Set H06.02 to 4 to define the speed reference in the speed control mode through H31.09 (unit: 0.001 rpm).

H31.11 Torque reference set through communication

Hexadecimal:	2031-0Ch	Effective Time:	Real time
Min.:	-100.000	Unit:	%
Max.:	100.000	Data Type:	Int32
Default:	0.000	Change:	Immediately

Value Range:

-100.000% to 100.000%

Description

Set H07.02 to 4 to define the torque reference in the torque control mode through H31.11 (unit: 0.001%). The setpoint 100.000% corresponds to the rated torque of the motor.